Supplementary Material for "Towards Trustworthy IoT Network Security: Interpretable Open-Set Intrusion Detection"

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This supplement provides the complete concept and rule definitions, the CTU-IoT-23 feature derivation, the threshold-gradient derivation, expanded ablations, per-class and per-family results, and additional robustness analyses. The main article presents the methods, principal evidence, and limitations supporting the central conclusions. Table, figure, and section numbers here are prefixed "S"; references to the main article use its numbering.

Feature set. Following the known-class feature-selection control (Section SIX), all CTU-IoT-23 results in the main article and in the finding-bearing supplementary tables (this control, the α-gate sweep and head-agreement, the per-family open-set breakdown, the concept-accuracy and leakage probe, the per-class table and the temporal split, the λ sensitivity and the (λ, γ) grid) use the leak-free known-class-only feature selection; the paired comparison with the originally submitted feature list is in Table S12. Every split is drawn deterministically by the released scripts, and the released code includes a script that prints every table value from the released results.

SI. The α-Gate and Interpretability Trade-Off

We sweep λ_α over six values, five seeds each (Table S2). The regulariser raises the learned gate *monotonically*. On CTU-IoT-23, α rises from 0.38 (λ_α=0) to 0.89 (λ_α=1.0); on CIC-IoT-2023, from 0.32 to 0.95. F1 does not move: the known-class *validation* F1 (the model-selection split reported in this sweep) stays near 0.933 on CTU and 0.818–0.819 on CIC; the corresponding held-out *test* F1 is 0.933 on CTU (Table III). OOD AUROC is flat on CTU (within 0.016 across the sweep) and rises on CIC, from 0.647 at λ_α=0 to 0.658–0.673 for every $\lambda_\alpha \geq 0.05$, a gain of 0.01–0.03 (up to about one seed std) that accompanies the increase in α; as the next paragraph shows, it is not evidence that the decision moved onto the rules.

Does raising α actually move the decision onto the rules? No. A larger gate weight need not mean more rule reliance, because the two heads can emit logits of different magnitudes and the learned weights can compensate. Testing reliance directly by branch agreement (Table S1), the deployed known-class prediction (before rejection) stays with the *neural-only* head at every λ_α, and its agreement with the *rule-only* head rises only mildly as α increases (CTU 32% to 51%), staying well below neural agreement. So λ_α tunes the gate scalar but not decision-level reliance on the rules; we describe α as the learned rule-head weight in the logit blend, not as a measure of how interpretable the decision is. (This is also a global scalar gate, so it cannot route individual hard flows to the neural branch.) Each point was trained from scratch.

TABLE S1: Branch agreement across the λ_α sweep (NeSy-NIDS, mean over five seeds): learned α and the fraction of flows on which the deployed known-class prediction (before rejection) equals the neural-only (dep=neural) or rule-only (dep=rule) prediction. The open-set rejection score is computed on the hard rule vector, so these figures do not measure the rules' influence on rejection.

λ_α	0	0.05	0.1	0.2	0.5	1.0
CTU α	0.38	0.58	0.69	0.81	0.86	0.89
CTU dep=neural	1.00	1.00	1.00	1.00	1.00	1.00
CTU dep=rule	0.32	0.36	0.38	0.48	0.51	0.48
CIC α	0.32	0.81	0.86	0.90	0.93	0.95
CIC dep=neural	0.99	0.98	0.97	0.98	0.98	0.97
CIC dep=rule	0.45	0.52	0.53	0.53	0.52	0.55

TABLE S2: α-gate sweep (NeSy-NIDS, mean over five seeds): learned α and OOD AUROC per λ_α.

λ_α	0	0.05	0.1	0.2	0.5	1.0
CTU α	0.38	0.58	0.69	0.81	0.86	0.89
CTU AUROC	0.904	0.900	0.909	0.900	0.899	0.893
CIC α	0.32	0.81	0.86	0.90	0.93	0.95
CIC AUROC	0.647	0.658	0.672	0.659	0.660	0.673

SII. Concept Leakage and the γ Ablation

Table S3 reports mean concept prediction accuracy for each CBM variant over five seeds. This quantity is the *concept accuracy* of the main text: agreement between the bottleneck's predicted concept and the declared threshold label, i.e. how faithfully the model reproduces the declared predicate, not whether the predicate is semantically "correct". JointCBM without concept regularisation (γ=0) reaches 87.4% on CTU and 96.3% on CIC. On CTU, accuracy rises to 94.2% at γ=1.0 (not monotonically: 89.1% at γ=0.1, 88.9% at 0.5); at γ=0 the weakest concept is `is_persistent` at 81% (it rises to 91% by γ=1.0). On CIC it rises from 96.3% to 97.2%, with every concept above 90%. SequentialCBM reaches near-perfect concept accuracy (99.9% / 99.5%) because its concept head is trained on the concept labels alone and frozen before the classifier is trained. HybridCBM is similar (99.6% / 99.2%):

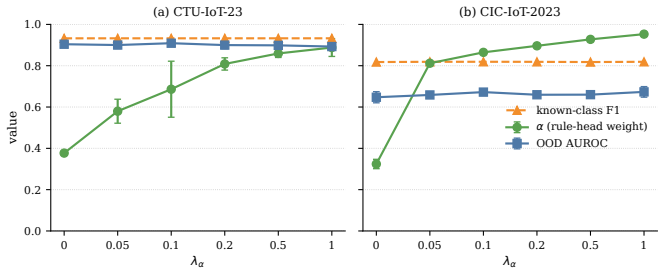


Fig. S1: α-gate sweep (NeSy-NIDS, mean over five seeds; error bars ± 1 std). Raising λ_α drives the learned gate α (the rule *weight* in the logit blend, green) monotonically upward while known-class F1 (orange, dashed) stays flat and OOD AUROC (blue) is flat on CTU-IoT-23 but rises modestly on CIC-IoT-2023.

TABLE S3: Mean concept prediction accuracy (%) on the known-class test split per CBM variant (mean over five seeds; λ sensitivity in Table S4).

Model	**CTU-IoT-23**	**CIC-IoT-2023**
JointCBM (γ=0)	87.4	96.3
JointCBM (γ=0.1)	89.1	96.5
JointCBM (γ=0.5)	88.9	96.8
JointCBM (γ=1.0)	94.2	97.2
SequentialCBM	99.9	99.5
HybridCBM	99.6	99.2

its skip connection lets the classifier bypass the bottleneck, so the concept head is under less classification pressure and can fit the concept labels more closely. Higher concept accuracy means closer agreement with the threshold labels; as the OOD results show, it does not by itself buy better detection.

Higher agreement with the labels does not by itself show that the bottleneck is free of *concept leakage*. We therefore train a linear classifier on the training split to predict the class from (i) the soft predicted concepts $\hat{\mathbf{c}} \in [0,1]^K$ and (ii) the same predictions thresholded at 0.5, and report the test-accuracy difference. This difference measures the additional predictive utility of the continuous values for these probes. It is small on CTU-IoT-23 for $\gamma \leq 0.5$ (five-seed means 0.001–0.013) but reaches $0.084 \pm .088$ at γ=1.0, where two of five seeds gain 0.17–0.21; on CIC-IoT-2023 it is 0.04–0.06 and grows rather than shrinks with γ (γ=0: $0.041 \pm .008$; γ=1.0: $0.058 \pm .008$).

A small difference does not establish semantic fidelity. Both probes use *predicted* concepts, and the thresholded predictions are not the declared predicates: where they disagree with the declared labels, the pattern of disagreement can itself carry class information. The predicate-replacement test of Section SXIX therefore uses the declared predicates directly.

SIII. Sensitivity to the Concept-Supervision Weight λ

Table S4 varies the concept-supervision weight λ for JointCBM (γ=0) on the main splits, with the training and evaluation flows of main Table III; the λ=0.5 column consists of the Table III models themselves. Known-class F1 is unchanged between λ=0.1 and 0.5 (validation F1 within 0.0005); at λ=1.0 one of five CTU seeds collapses (test F1 0.734), lowering the CTU mean to 0.893. Raising λ raises concept accuracy on both datasets and moves OOD AUROC in opposite directions, by $+0.020$ on CTU-IoT-23 and -0.040 on CIC-IoT-2023 from λ=0.1 to 1.0.

TABLE S4: Sensitivity to the concept-supervision weight λ (JointCBM, γ=0, main splits, mean over five seeds; AUROC ± 1 std). The λ=0.5 column is the JointCBM (γ=0) row of main Table III.

	λ=0.1	λ=0.5	λ=1.0
CTU F1	0.9331	0.9330	0.8931
CTU concept-acc	0.700	0.874	0.935
CTU OOD AUROC	0.810±0.021	0.823±0.044	0.830±0.041
CIC F1	0.8196	0.8183	0.8184
CIC concept-acc	0.865	0.963	0.975
CIC OOD AUROC	0.743±0.018	0.717±0.017	0.703±0.013

SIV. Temporal (Timestamp-Ordered) Split on CTU-IoT-23

Random flow partitioning can leak device or session identity between train and test. The CTU attacks come from very few source addresses (Okiru: 4, DDoS: 7, C&C-HeartBeat: 5); Section SV holds out source IP addresses, and here we build the known-class splits in time: we take the same per-class sample as the main split, order the flows of each known class by capture timestamp (timestamps are compared only within a class), and assign that class's earliest 70% to training, the next 15% to validation and the latest 15% to test (five seeds; within each class, test strictly follows train; the held-out unknown families are unchanged). Timestamp ordering does not by itself guarantee a session- or capture-disjoint split, individual sessions or captures may straddle the train/test boundary, so we claim temporal ordering, not capture-, session- or device-disjointness. A capture-disjoint split needs the capture identifier, which our merged flow table does not keep, and remains untested. This protocol probes a property of the CTU-IoT-23 labelling: the *DDoS* family is temporally bimodal, a high-rate flood capture (September 2018, median Rate $\approx 9.3 \times 10^5$ pps) and near-zero-rate captures (December 2018 onward, median Rate <1), so a timestamp split can train on one DDoS regime and test on the other. This makes *every* model score F1 0.00 on DDoS on every seed; with the main split's 31-feature list on the same sample only two JointCBM γ=0.5 seeds keep DDoS (F1 0.65), so the collapse follows from the label's temporal structure rather than from feature selection (Section SIX).

Two effects separate. The *absolute* deterioration is large but comes from one class: four-class weighted F1 falls from 0.933 (random split) to 0.646–0.648 for every model but SequentialCBM ($F1_4$, main Table IV) because DDoS is lost; C&C-HeartBeat also falls, from 0.998 to about 0.79. This is a temporal-generalisation cost, not an interpretability one. The *between-model* gap is negligible: on the three coherent families every JointCBM variant, HybridCBM and NeSy-NIDS match the MLP in aggregate (Coh-3 0.862–0.863) while redistributing

errors across classes (e.g. γ=0: Benign 0.841 vs. 0.958, Okiru 0.961 vs. 0.844). SequentialCBM is the exception, losing Benign ($F1_4$ 0.330, Coh-3 0.441). NeSy-NIDS gives the highest temporal-split AUROC (0.888±0.025; 0.901±0.030 with the α-gate penalty of Section SI), and the JointCBM γ ordering shifts relative to the alternative split.

SV. Source-IP Holdout

Flows are grouped by source IP address. Our merged flow tables keep the source IP but not the capture name, and an address can recur across captures, so a held-out address is not shown to be a single physical device; the experiment is a source-IP holdout, and its conclusions concern source addresses. Two CTU addresses, 192.168.1.198 and 192.168.100.111, have at least 3,450 flows in each of the four known classes. For each, we take that address's flows as the known test set (3,450 per class) and draw the training and validation flows, in the usual numbers, from all other addresses; the unknown test flows are those of the main split. The feature list is re-selected on each split's training rows (36 features for address .198, 33 for address .111), and a control uses the main split's 31-feature list on the same rows. The MLP, the CBM variants (JointCBM, SequentialCBM, HybridCBM) and NeSy-NIDS (with the α penalty and the rule-only head) are trained with five seeds and scored on the shared evaluation flows; for address .198 one class has only 14,699 training flows instead of 16,100, because C&C-HeartBeat has fewer flows on the other addresses. `06_make_kfs_splits.py -split device` builds the splits (`ctud_h198`, `ctud_h111` and the `_o` controls).

TABLE S5: CTU source-IP holdout: known-class weighted F1 (five-seed mean) and OOD AUROC (mean±std, five seeds) with the re-selected features, and AUROC with the main 31-feature list (31f). With the 31-feature list, F1 is 0.417 on address .198 (0.517 for NeSy-NIDS (α)) and 0.149–0.227 on address .111. Main split for reference: F1 0.566–0.933, AUROC 0.805–0.909.

	Address 192.168.1.198			Address 192.168.100.111		
Model	F1	AUROC	AUROC 31f	F1	AUROC	AUROC 31f
MLP baseline	0.417	0.528±.020	0.529±.013	0.228	0.219±.045	0.383±.015
JointCBM γ=0	0.417	0.738±.113	0.824±.081	0.223	0.211±.102	0.111±.048
JointCBM γ=0.1	0.417	0.809±.073	0.770±.084	0.214	0.235±.085	0.113±.043
JointCBM γ=0.5	0.417	0.784±.050	0.816±.083	0.200	0.256±.088	0.114±.029
JointCBM γ=1.0	0.417	0.819±.048	0.789±.062	0.196	0.161±.015	0.093±.028
SequentialCBM	0.417	0.865±.001	0.866±.002	0.192	0.170±.013	0.202±.030
HybridCBM	0.417	0.821±.033	0.822±.041	0.152	0.215±.038	0.230±.042
NeSy-NIDS	0.417	0.843±.042	0.846±.045	0.161	0.284±.043	0.279±.044
NeSy-NIDS (α)	0.417	0.847±.045	0.847±.045	0.136	0.278±.058	0.280±.043
Rule-only	0.416	0.770±.032	0.770±.032	0.177	0.193±.035	0.193±.035

Known-class classification fails for every model. The held-out addresses' class traffic is unlike the other addresses': none of address .198's Benign or DDoS test flows, and none of address .111's test flows in any class, has an exact duplicate among the training flows of its class (Section SV-A), and address .198's 3,450 Benign test flows are three distinct feature vectors. With the re-selected features, every MLP, CBM and NeSy-NIDS run on address .198 scores F1 0.00 on Benign and on one more class (DDoS, or C&C-HeartBeat for SequentialCBM and rule-only), which fixes weighted F1 at 0.417 (with the 31-feature list, two NeSy-NIDS (α) seeds reach 0.67 on Benign); the MLP (seed 0) classifies 3,448 of the 3,450 Benign flows as Okiru and all DDoS flows as Benign. On address .111 every MLP, CBM and NeSy-NIDS run with the re-selected features scores F1 below 0.03 on DDoS and Okiru (with the 31-feature list SequentialCBM reaches 0.54 on DDoS); the MLP (seed 0) classifies 3,407 of the DDoS and 3,409 of the Okiru flows as C&C-HeartBeat. Because the known test flows are themselves unlike the training data, AUROC no longer isolates novel attack types. On address .111 it is below 0.5 for every model and seed (all 100 runs): the held-out address's known flows score as more novel than the unknown flows, over 99.9% of which come from other addresses that also produce known-class flows (89% for address .198, where 3,516 unknown flows come from the held-out address itself). On address .198 SequentialCBM and NeSy-NIDS score highest and the MLP lowest, but with every model losing two of the four known classes these values do not measure open-set detection on a new address.

A. A Third CTU Address and TON_IoT

Address 192.168.100.108 has 3,450 flows in each CTU attack class but only 3,051 Benign flows, all of which form its Benign test set; it has no unknown-family flows. On TON_IoT the attack flows of each known class come from a few source addresses; `07_make_splits_ton.py -holdout-attacker` takes 7,500 test flows per attack class from address 192.168.1.30 or 192.168.1.31 and 7,500 normal flows from address 192.168.1.195 or 192.168.1.190, and draws training and validation flows from all other addresses (`tond_a30`, `tond_a31`; the fixed TON_IoT feature list, scaler and vocabularies refit on the training rows). NeSy-NIDS has no TON_IoT rule bank.

TABLE S6: CTU address .108 and the two TON_IoT source-IP holdouts: known-class weighted F1 (five-seed mean) and OOD AUROC against the main split's unknown pool (mean±std, five seeds). 31f: the main 31-feature CTU list. TON_IoT main split: F1 0.704–0.898, AUROC 0.552–0.654.

	CTU address .108			TON_IoT address .30		TON_IoT address .31	
Model	F1	AUROC	AUROC 31f	F1	AUROC	F1	AUROC
MLP baseline	0.544	0.548±.003	0.542±.008	0.761	0.552±.024	0.886	0.493±.033
JointCBM γ=0	0.541	0.600±.030	0.590±.122	0.754	0.635±.007	0.885	0.585±.012
JointCBM γ=0.1	0.539	0.577±.056	0.554±.071	0.766	0.659±.027	0.853	0.597±.026
JointCBM γ=0.5	0.547	0.595±.068	0.625±.037	0.746	0.664±.023	0.858	0.613±.033
JointCBM γ=1.0	0.546	0.632±.059	0.635±.050	0.753	0.660±.017	0.888	0.617±.011
SequentialCBM	0.343	0.753±.049	0.729±.031	0.677	0.596±.009	0.731	0.601±.016
HybridCBM	0.544	0.620±.049	0.689±.077	0.767	0.610±.016	0.892	0.598±.020
NeSy-NIDS	0.552	0.907±.011	0.899±.010	–	–	–	–
NeSy-NIDS (α)	0.541	0.906±.014	0.898±.010	–	–	–	–
Rule-only	0.475	0.887±.011	0.887±.011	–	–	–	–

On TON_IoT the largest per-class losses are on the held-out normal address and on password (address .30, MLP: normal 0.68 vs. 0.91, password 0.59 vs. 0.79 on the main split); address .31 matches the main split. The share of known test flows with an exact duplicate (features rounded to four decimals) among the training flows of their class is 0.84 on the CTU main split and 0.88 on TON_IoT, 0.42, 0.00 and 0.41 on CTU addresses .198, .111 and .108, and 0.74 and 0.77 on TON_IoT addresses .30 and .31.

B. Open-Set Test with Unknown Flows from the Held-Out Address

Table S5 scores against the main split's unknown pool, most of which comes from addresses seen in training. `cbm/run_device_openset.py` instead scores each held-out address's known test flows against that address's own unknown-family flows, for the MLP, CBMs and NeSy-NIDS: address .198 (3,516: C&C 3,499, C&C-FileDownload 14, Okiru-Attack 3) and the DoS flows of TON_IoT address .30 (with normal address .195) and address .31 (with .190), 2,000 each, sampled as in the main split. The MITM flows of the two TON_IoT addresses (2 and 510) are scored separately and are not part of Tables S7–S8. As a seen-address reference, the main-split models are scored the same way against the main split's own C&C (CTU, 10,000 of 21,995 flows) or DoS (TON_IoT, 2,000) flows. Address .111 has 4 unknown-family flows and address .108 none, so neither is tested.

Two operating points are reported. TPR@5%FPR (Table S7), as in main Table III, thresholds at the 95th percentile of the held-out address's own known test scores: it is a point on the test ROC curve, a diagnostic that is not available before deployment. Table S8 uses the deployable threshold τ of main Section III-D, the 95th percentile of the known validation scores, whose flows come from the training addresses, and reports the share of the held-out address's known test flows it rejects and the recall of the address's unknown flows.

TABLE S7: OOD AUROC / diagnostic TPR@5%FPR (five-seed mean) with known and unknown test flows from the same source address. "Seen": main-split models against the same family's flows from addresses seen in training. TON_IoT columns use DoS flows only.

	CTU, C&C			TON_IoT, DoS		
Model	Seen	Address .198	Address .198, 31f	Seen	Address .30	Address .31
MLP baseline	0.853 / 0.337	0.396 / 0.007	0.396 / 0.007	0.444 / 0.009	0.411 / 0.004	0.818 / 0.016
JointCBM γ=0	0.829 / 0.339	0.609 / 0.052	0.713 / 0.162	0.754 / 0.016	0.691 / 0.007	0.749 / 0.016
JointCBM γ=0.1	0.832 / 0.331	0.663 / 0.071	0.688 / 0.066	0.735 / 0.011	0.705 / 0.003	0.757 / 0.024
JointCBM γ=0.5	0.843 / 0.439	0.637 / 0.020	0.712 / 0.100	0.725 / 0.014	0.750 / 0.002	0.792 / 0.026
JointCBM γ=1.0	0.854 / 0.507	0.690 / 0.073	0.665 / 0.030	0.735 / 0.011	0.703 / 0.003	0.772 / 0.001
SequentialCBM	0.878 / 0.497	0.635 / 0.039	0.638 / 0.041	0.719 / 0.022	0.693 / 0.000	0.691 / 0.002
HybridCBM	0.876 / 0.554	0.636 / 0.033	0.637 / 0.032	0.722 / 0.021	0.690 / 0.001	0.645 / 0.011
NeSy-NIDS	0.865 / 0.260	0.651 / 0.060	0.651 / 0.060	–	–	–
NeSy-NIDS (α)	0.872 / 0.247	0.651 / 0.066	0.650 / 0.057	–	–	–
Rule-only	0.820 / 0.183	0.639 / 0.031	0.639 / 0.031	–	–	–

TABLE S8: Deployable threshold τ (95th percentile of known validation scores): share of known test flows rejected / recall of unknown flows (five-seed mean). The target rejection rate is 5%.

	CTU, C&C			TON_IoT, DoS		
Model	Seen	Address .198	Address .198, 31f	Seen	Address .30	Address .31
MLP baseline	0.049 / 0.337	0.501 / 0.049	0.501 / 0.049	0.050 / 0.009	0.070 / 0.004	0.131 / 0.070
JointCBM γ=0	0.048 / 0.276	0.250 / 0.028	0.051 / 0.038	0.052 / 0.016	0.062 / 0.008	0.060 / 0.016
JointCBM γ=0.1	0.051 / 0.335	0.151 / 0.027	0.201 / 0.033	0.051 / 0.016	0.057 / 0.003	0.081 / 0.037
JointCBM γ=0.5	0.052 / 0.439	0.200 / 0.035	0.151 / 0.037	0.051 / 0.014	0.060 / 0.005	0.058 / 0.047
JointCBM γ=1.0	0.053 / 0.513	0.126 / 0.131	0.200 / 0.033	0.051 / 0.017	0.070 / 0.006	0.077 / 0.003
SequentialCBM	0.055 / 0.549	0.000 / 0.031	0.000 / 0.032	0.051 / 0.021	0.099 / 0.003	0.062 / 0.002
HybridCBM	0.050 / 0.552	0.050 / 0.026	0.100 / 0.027	0.051 / 0.021	0.088 / 0.004	0.037 / 0.002
NeSy-NIDS	0.047 / 0.286	0.030 / 0.042	0.022 / 0.041	–	–	–
NeSy-NIDS (α)	0.047 / 0.268	0.016 / 0.040	0.023 / 0.040	–	–	–
Rule-only	0.039 / 0.183	0.177 / 0.045	0.177 / 0.045	–	–	–

On CTU the held-out address lowers C&C AUROC from 0.82–0.88 to 0.40–0.71 and diagnostic TPR from 0.18–0.55 to at most 0.162. On TON_IoT address .30 the change is small (AUROC 0.41–0.75 vs. 0.44–0.75), and TPR is low for every model on seen addresses too (≤ 0.022). With the deployable threshold the known rejection rate stays near its target on seen addresses (3.9–5.5%) but not on held-out ones (0–50% on address .198, 3.7–13.1% on the TON_IoT addresses), and unknown recall is at most 13.1% (.198) and 7.0% (TON_IoT). The ordering of the models is not stable across addresses: the concept models are above the MLP on address .198 and address .30, and below it on address .31 (MLP 0.818). On the 510 MITM flows of address .31, AUROC is 0.63–0.85 and diagnostic TPR at most 0.139.

SVI. Confidence Intervals

`cbm/run_bootstrap_ci.py` recomputes the per-flow OOD scores of every model compared from its saved checkpoints (checked against the stored AUROC) and draws 2,000 hierarchical bootstrap replicates: the five seeds are resampled with replacement for each model independently, and the 5,000 known and 5,000 unknown evaluation flows are resampled with replacement and shared by the two models of a comparison, since every model is scored on the same flows. The interval is the 2.5–97.5 percentile of the difference of the two seed-mean AUROCs. The intervals cover the variation from training seeds and from the sampled evaluation flows under the chosen splits; they do not measure uncertainty across new captures, source addresses or independently drawn splits.

TABLE S9: Differences in mean OOD AUROC with 95% bootstrap confidence intervals.

Data	Comparison	Difference	95% CI
CTU	NeSy-NIDS − MLP	+0.030	[+0.016, +0.043]
CTU	NeSy-NIDS (α) − MLP	+0.035	[+0.020, +0.047]
CTU	SequentialCBM − MLP	−0.002	[−0.027, +0.018]
CTU	JointCBM γ=1.0 − JointCBM γ=0	+0.011	[−0.052, +0.065]
CTU	JointCBM γ=0 − 8-unit bottleneck	+0.263	[+0.102, +0.380]
CIC	MLP − JointCBM γ=0	+0.090	[+0.073, +0.108]
CIC	JointCBM γ=1.0 − JointCBM γ=0	−0.018	[−0.037, +0.002]
CIC	8-unit bottleneck − JointCBM γ=0	+0.003	[−0.024, +0.027]
CIC	NeSy-NIDS, energy score − NeSy-NIDS, Mahalanobis	+0.073	[+0.035, +0.112]
TON	JointCBM γ=0 − MLP	+0.098	[+0.051, +0.145]
TON	JointCBM γ=0 − 8-unit bottleneck	+0.106	[+0.004, +0.214]

Every difference the main text describes as a gap excludes zero. An interval that includes zero shows that this analysis does not resolve the difference, not that the two models are equivalent; the main text describes those comparisons as similar observed AUROC or as no consistent improvement. The TON_IoT difference between JointCBM and the 8-unit bottleneck barely excludes zero, because the bottleneck varies widely across seeds.

SVII. Rule Selectivity and Threshold Drift

All values in this section are mean binarised activations (Eq. 5) per known class on the known-class validation split for the seed-0 model, the same data as Fig. 2. On CTU-IoT-23, several rules fire selectively but not deterministically (`cc_heartbeat_beaconing` at 0.78 and `incomplete_handshake` at 0.87 on C&C-HeartBeat, `asymmetric_flood` at 0.81 on DDoS and 0.13 on Benign, `ddos_high_rate` at 0.70 on DDoS and 0.21 on Okiru), leaving within-class variance for Mahalanobis to work with; others fire across several classes (`okiru_low_rate`: 0.79

on Okiru but 0.87 on Benign and 0.61 on C&C-HeartBeat). On CIC-IoT-2023 firing is near-deterministic (`syn_flood_ddos` at 1.00 on DDoS-SYN, `benign_low_rate` at 0.97 on Benign but also 0.91 on Mirai), within-class variance is close to zero, and Mahalanobis has little to separate on (AUROC 0.647). These two activation regimes coincide with the observed CTU-versus-CIC difference in Mahalanobis AUROC, but, as the main paper establishes, known-class activation geometry does not by itself govern detectability, so we report the pattern as descriptive, not causal (Fig. 2). The learned thresholds move away from their expert initial values (seed 0): on CTU-IoT-23 the `cc_heartbeat_beaconing` bounds move by up to -0.50, and on CIC-IoT-2023 by up to -0.88 on `syn_flood_ddos`; across the five seeds the per-condition drift vectors are correlated on CIC-IoT-2023 (mean pairwise r=0.73) but not on CTU-IoT-23 (r=0.05; script `nesy/run_fidelity.py`), so we read the CIC refinement as systematic and the CTU pattern as indicative only (main Fig. 3).

SVIII. Configuration Sensitivity of the Geometry Correlations

Main Table XIII correlates each known-class statistic with OOD AUROC over 30 CBM models per dataset, which are five seeds of six configurations rather than 30 independent observations. Table S10 reports, on the same leak-free feature lists as the main table (CTU: known-class-only selection, Section SIX; CIC/TON: fixed lists), two further aggregations applied identically to all three corpora: the correlation over the six configuration means (r_6, n=6), and the range of the 30-model correlation when each configuration in turn is excluded (leave-one-configuration-out, LOCO). On CIC-IoT-2023 and TON_IoT the seven statistics agree in direction, and the six separability and variance statistics are strongly associated with AUROC at the configuration-mean level ($|r_6| \geq 0.93$ on both); closed-set F1 is weaker (r_6=0.63 and 0.73) and is the only statistic whose sign changes under LOCO there (on CIC). CTU-IoT-23 is different. All seven statistics take the opposite sign (r_6 from -0.82 to $+0.49$), and the LOCO ranges show why: every CTU range reaches its bound closest to zero when SequentialCBM is excluded, where the correlations fall to $|r| \leq 0.22$ and Fisher ratio, silhouette and closed-set F1 change sign. SequentialCBM combines the least separable concept space and the lowest closed-set F1 with one of the highest CTU AUROCs, so this single configuration sets the CTU direction. The unknown/known ratio, which uses the unknown flows, is positive everywhere (r_6 0.70–0.92). These are associations with Mahalanobis AUROC measured *within* each corpus, and the strong r_6 values rest on six points. Prediction on a held-out corpus and scorer selection are tested separately (Sections SXIV and SXV), and neither succeeds. The LOCO analysis thus locates the CTU reversal in one configuration rather than in the corpus.

TABLE S10: Configuration sensitivity of main Table XIII, on the same leak-free feature lists (CTU: known-class-only selection). r_6: Pearson r over the six configuration means. LOCO: range of the 30-model r when one configuration is excluded; on CTU the bound nearest zero is always the run without SequentialCBM. *Sign changes when SequentialCBM is excluded.

	CTU-IoT-23		CIC-IoT-2023		TON_IoT	
Statistic	r_6	LOCO	r_6	LOCO	r_6	LOCO
Within-class variance	$+0.49$	$[+0.01, +0.42]$	-0.96	$[-0.82, -0.66]$	-0.95	$[-0.54, -0.32]$
Fisher ratio	-0.45	$[-0.32, +0.02]^*$	$+0.96$	$[+0.56, +0.76]$	$+0.95$	$[+0.36, +0.54]$
Ridge-LDA	-0.50	$[-0.44, -0.08]$	$+0.95$	$[+0.31, +0.60]$	$+0.98$	$[+0.40, +0.57]$
Silhouette	-0.46	$[-0.42, +0.01]^*$	$+0.95$	$[+0.74, +0.85]$	$+0.94$	$[+0.40, +0.58]$
Nearest-centroid margin	-0.47	$[-0.43, -0.02]$	$+0.94$	$[+0.50, +0.74]$	$+0.94$	$[+0.32, +0.52]$
Concept entropy	-0.52	$[-0.47, -0.06]$	$+0.96$	$[+0.71, +0.83]$	$+0.93$	$[+0.25, +0.49]$
Closed-set F1	-0.82	$[-0.42, +0.22]^*$	$+0.63$	$[-0.28, +0.76]^*$	$+0.73$	$[+0.03, +0.42]$
Unknown/known ratio	$+0.78$	$[+0.59, +0.74]$	$+0.92$	$[+0.43, +0.64]$	$+0.70$	$[+0.70, +0.76]$

SIX. CTU-IoT-23 Feature Derivation

CTU-IoT-23 is distributed as packet captures with labelled Zeek `conn.log` files. For each capture we compute per-flow packet statistics with DPKT (field names follow the CIC-IoT-2023 feature set) and join each labelled Zeek record to the DPKT flows with the same source and destination IP, ports and protocol whose start time lies in $[t_{\text{zeek}}-\text{duration}-60\,\text{s},\ t_{\text{zeek}}+60\,\text{s}]$, keeping the one closest in start time (`preprocess/merge_iot23_outer2.py`), so each Zeek record yields at most one row; in the stored table only two pairs of rows share source address, timestamp, duration, byte counts, history and label. Zeek records without a DPKT match are kept, with their DPKT features set to 0: 1,896,696 of the 110,879,826 retained flows (1.7%), mostly DDoS (7.6% of DDoS flows); no flow matched with protocol ICMP. In the main split they are 2.2% of the training and 2.0% of the known test flows (mostly DDoS) and 0.4% of the unknown flows. Before sampling, a label filter excludes PartOfAHorizontalPortScan, PartOfAHorizontalPortScan-Attack and C&C-PartOfAHorizontalPortScan, so horizontal port scanning is neither a known nor a held-out class. We used the full IoT-23 archive (`iot_23_datasets_full.tar.gz`), whose packet captures the DPKT features require, and keep the labels of its Zeek label files. For capture CTU-IoT-Malware-Capture-33-1 these files label 13,609,467 flows Okiru, a count the provider's summary table lists under Okiru-Attack; our flows from that capture's source address (192.168.1.197) include exactly 13,609,467 Okiru and no Okiru-Attack flows, and the label file for this capture currently served by the provider also uses Okiru. These flows therefore belong to the known Okiru class, and our three Okiru-Attack flows are those of capture 36-1; Table S11 lists the retained labels (`checks/ctu_label_counts.py`).

Four ratios are derived from Zeek fields: byte ratio $\text{orig_bytes}/(\text{resp_bytes}+1)$, originator packet and byte rates $\text{orig_pkts}/(\text{duration}+1)$ and $\text{orig_ip_bytes}/(\text{duration}+1)$, and direction $\text{orig_bytes}-\text{resp_bytes}$. The model uses 34 numeric features (Zeek: duration, orig/resp bytes, packets and IP bytes; DPKT: Rate, Srate, IAT, Number, Tot_sum, Min, Max, AVG, Std, Variance, Magnitude, Radius, ack/syn/fin/rst counts, fin/rst flag numbers, and HTTP, HTTPS, SSH, IRC, Telnet indicators; plus the four derived ratios) and three Zeek categorical fields (conn_state, history, service) integer-encoded, 37 in total. Missing values are set to -1; features are not rescaled.

TABLE S11: Retained CTU-IoT-23 flows per label and those without a matching DPKT flow (DPKT features set to 0).

Label	Retained	No DPKT match
Okiru	60,717,929	267,008
Benign	30,830,238	158,334
DDoS	19,265,778	1,471,222
C&C-HeartBeat	33,537	6
C&C	21,995	103
Attack	9,398	1
C&C-HeartBeat-Attack	834	18
C&C-FileDownload	53	4
C&C-Torii	30	0
FileDownload	18	0
C&C-HeartBeat-FileDownload	11	0
Okiru-Attack	3	0
C&C-Mirai	2	0

Feature-selection caveat. The 34 numeric features were chosen before this study as those covering 99% of cumulative random-forest importance, with a class-weighted forest trained on up to 10^6 flows of every labelled CTU family. That sample included the nine families later held out as unknown, so the CTU feature set is not blind to the held-out labels. The choice is fixed and shared by every model compared, but it may favour separability of the CTU unknowns in raw-feature space. CIC-IoT-2023 and TON_IoT use fixed feature lists (the 39 DPKT columns; 23 conn-log fields) without label-informed selection.

Known-class feature-selection control. Disclosing this is not enough, so we ran the control. We re-selected the CTU numeric features with the same random-forest procedure but on the known-class *training* rows only, with the nine held-out families excluded from selection. The known-only list keeps 28 numeric features (of 34): it drops the service and protocol indicators (HTTP, HTTPS, SSH, IRC), two counters (`resp_pkts`, `fin_count`) and several statistical or derived features (Radius, Std, Variance, `resp_ip_bytes`, `rst_flag_number`), and adds TCP, Protocol_Type, Time_To_Live and two TCP-flag indicators (`cwr`, `syn` flag numbers), 31 features in total with the three categorical fields. We drew a class-balanced split and ran every CTU comparison with this list (`ctuk`); as a paired baseline we ran the submitted 37-feature list on the *same* rows (`ctuko`), so `ctuko` vs `ctuk` isolates the selection source from the sample. The NeSy-NIDS rule features are expert-defined rather than RF-selected, so the two rule counters the known-only forest drops are restored and the rule indices remapped; only the neural branch and the CBM/MLP/DT/RF inputs change. The script `06_make_kfs_splits.py` and the drivers `cbm/run_kfs_control.py` and `nesy/run_kfs_control.sh` reproduce the table.

Table S12 reports the outcome over five seeds. The accuracy and scorer findings hold. The JointCBM variants, HybridCBM and NeSy-NIDS still match the MLP on known-class F1 (all 0.933); the two low-capacity models are unchanged (SequentialCBM 0.57, post-hoc CBM 0.58). Per-class Mahalanobis is the best CTU scorer for every architecture under both lists (Table S13), with the logit-energy score below chance for the MLP (0.39 vs. Mahalanobis 0.88). Concept-fidelity regularisation γ still has no monotone effect on OOD AUROC (JointCBM 0.82–0.83 and 0.81–0.84 across γ; Table S12). NeSy-NIDS, the strongest CTU detector, moves from 0.885 to 0.904.

Three OOD AUROC changes reach the 0.05 level read as material elsewhere, and all are gains under known-class-only selection. SequentialCBM rises from $0.782\pm.042$ to $0.872\pm.026$ ($\Delta = +0.090$, beyond its seed std), HybridCBM from $0.772\pm.073$ to $0.822\pm.055$ ($\Delta = +0.050$, below its own seed std) and Post-hoc CBM from 0.805 to 0.871; the MLP moves -0.020, the JointCBM variants by at most 0.03, and the DT and RF baselines gain 0.12 and 0.03. Removing the held-out labels from feature selection therefore does not flatter the CTU detection results; if anything the submitted list hurt the concept-space detectors.

The control also bears on the geometry reading. The CTU correlations of main Table XIII are opposite in sign to CIC-IoT-2023 and TON_IoT, and Section SVIII traces the direction to SequentialCBM, the CBM configuration whose AUROC changes most with the feature list. Under the leak-free list no CTU association holds without that configuration, and Section SXV shows that no statistic predicts detectability consistently on a held-out corpus.

Per-split selection for the robustness experiments. The known-only selection above is fitted on the primary random split's known-class training rows, which is not automatically leak-free for the other CTU experiments. We therefore run the same random-forest selection separately on each: the alternative known set's training rows (excluding the now-held-out C&C-HeartBeat), the earliest 70% of the timestamp-ordered split, and each leave-one-family-out fold's reduced known set, so the family that experiment holds out, and any test-period flow, never informs its features. The alternative split draws its own class-balanced sample, the temporal split reorders the main split's sample by time, and the leave-one-family-out folds reuse the main split's rows. For the alternative and temporal splits, running the main split's 31-feature list on the same rows separates re-selection from the sample, and the accuracy, scorer and γ conclusions hold under both. One consequence is visible in the temporal split (Section SIV): the DDoS collapse occurs under both feature lists (bar two JointCBM seeds with the main-split list), so it is a property of the temporal split, not of feature selection.

SX. NeSy-NIDS Rule Bank and Initial Thresholds

Table S14 gives the complete eight-rule bank for each dataset used by NeSy-NIDS (Section III-C of the main paper lists only representative rules). Each rule is a conjunction of threshold conditions on named flow features; the thresholds shown are the expert *initial* values $\boldsymbol{\theta}^{(0)}$, which are then refined by gradient descent through the k-annealed gates. The full feature lists (per-dataset `feature_cols`), the deterministic concept/rule threshold values, and the train/val/test split indices are released with the code (see the code-availability statement), so every representation in the paper is reproducible.

SXI. Threshold Gradient

Training uses the soft activations of main Eq. 4 throughout, so every threshold receives a gradient. For a greater-than

TABLE S12: Known-class feature-selection control on CTU-IoT-23 (five seeds). `ctuko`: split with the submitted 37-feature list; `ctuk`: same split with features selected from known-class training rows only, so `ctuko` vs `ctuk` isolates the selection source from the sample. Known-class weighted F1 (mean) and per-class Mahalanobis OOD AUROC (mean±std). Accuracy is unchanged; known-class-only selection raises the OOD AUROC of SequentialCBM (+0.090, beyond its seed std), HybridCBM (+0.050) and Post-hoc CBM (+0.066).

	F1		OOD AUROC	
Model	`ctuko`	`ctuk`	`ctuko`	`ctuk`
MLP	0.933	0.933	0.894±.005	0.874±.002
JointCBM (γ=0)	0.933	0.933	0.817±.031	0.823±.044
JointCBM (γ=0.1)	0.933	0.933	0.821±.033	0.805±.035
JointCBM (γ=0.5)	0.933	0.933	0.821±.030	0.844±.036
JointCBM (γ=1)	0.933	0.933	0.833±.029	0.834±.050
SequentialCBM	0.570	0.567	0.782±.042	0.872±.026
HybridCBM	0.933	0.933	0.772±.073	0.822±.055
NeSy-NIDS	0.933	0.933	0.885±.020	0.904±.014
Decision tree	0.934	0.934	0.557±.000	0.679±.000
Random forest	0.934	0.934	0.630±.002	0.657±.034
Post-hoc CBM	0.576	0.576	0.805±.030	0.871±.024

TABLE S13: Scorer comparison under the known-only feature selection (`ctuk`): OOD AUROC (mean±std, five seeds) for per-class Mahalanobis on the concept space, the logit-space energy score, and their average after z-normalising each with known validation scores (combined). Per-class Mahalanobis is best for every architecture, and the logit-energy score reaches at most 0.61 (main Table VII gives the CIC comparison, where the ranking depends on architecture).

Architecture	Mahalanobis	Energy	Combined
MLP	**0.879±.001**	0.386±.073	0.682±.046
JointCBM (γ=0)	**0.830±.044**	0.476±.111	0.631±.068
JointCBM (γ=0.1)	**0.812±.034**	0.494±.136	0.624±.098
JointCBM (γ=0.5)	**0.851±.037**	0.570±.138	0.714±.106
JointCBM (γ=1)	**0.842±.049**	0.612±.124	0.708±.100
SequentialCBM	**0.875±.026**	0.604±.085	0.757±.018
HybridCBM	**0.827±.055**	0.365±.047	0.610±.047

condition $i \in \mathcal{I}_r^{>}$ of rule r,

$$\frac{\partial \mathcal{L}}{\partial \theta_i} = \frac{\partial \mathcal{L}}{\partial a_r} \cdot \Big(-k(t)\, g_i^{>}(1-g_i^{>}) \prod_{j \neq i} g_j\Big), \qquad (1)$$

where $g_i^{>}=\sigma(k(t)(x_i-\theta_i))$ and g_j are the other gates of the rule; the sign flips for less-than conditions. As $k(t)$ grows, $g_i^{>}(1-g_i^{>})$ vanishes for gates far from their threshold, so late in training only borderline conditions keep moving. The straight-through estimator is not used in training.

SXII. OOD AUROC versus Closed-Set F1

Figure S2 plots every model of main Table III in the F1–AUROC plane; main Section IV-D discusses what it shows for the closed-set/open-set relationship.

SXIII. Scorer Settings

Table S15 lists how each OOD score is computed. Every setting was fixed in advance (the ODIN and kNN values are the published defaults) and none was chosen using the held-out families; scorers are fitted on training vectors, and the combined score is normalised with known validation scores, so no test flow enters any scorer.

TABLE S14: Complete NeSy-NIDS rule bank (R=8 per dataset): target class and the conjunction of conditions at the expert *initial* thresholds (refined during training). CTU-IoT-23 in raw units; CIC-IoT-2023 in z-scored units.

Rule	Target	Condition (initial threshold)
CTU-IoT-23 (raw units)		
ddos_high_rate	DDoS	Rate>1000
cc_heartbeat_beaconing	C&C-HB	1.5<IAT<5.0 ∧ 0.1<Rate<5.0
benign_single_probe	Benign	orig_pkts<2∧ Rate<100
okiru_low_rate	Okiru	Rate<500∧ duration<2.5
asymmetric_flood	DDoS	resp_pkts<1∧ orig_pkts>1
persistent_connection	C&C-HB	duration>2.5
incomplete_handshake	C&C-HB	syn_count>2∧ fin_count<1
ddos_short_burst	DDoS	Rate>1000∧ duration<0.1
CIC-IoT-2023 (z-scored units)		
syn_flood_ddos	DDoS-SYN	rate>0.15∧ syn_count>1.5
udp_mirai_flood	Mirai	tcp<−1.0∧ rate>0.0
recon_port_scan	Recon	rst_count>1.5
vuln_scan	VulnScan	fin_count>1.0∧ rate>0.0
benign_low_rate	Benign	rate<0.0∧ syn_count<0.0
large_payload_flood	Benign	avg>0.4∧ udp>0.0
high_entropy_scan	Recon	variance>0.5∧ rst_count>0.5
short_connection_flood	DDoS-SYN	iat<−0.003∧ rate>0.1

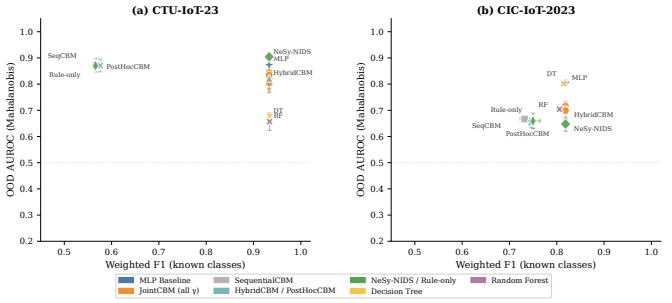


Fig. S2: OOD AUROC vs. known-class weighted F1 for all models on CTU-IoT-23 (left) and CIC-IoT-2023 (right), using the values of main Table III. Error bars are ±1 std over five seeds; every model uses five seeds (DT's raw-feature score has no seed variance).

SXIV. Leave-Family-Out Scorer Selection

The measured associations between known-class statistics and OOD AUROC vary across datasets (main Table XIII); whether a scorer can be *selected* in advance is a separate question, which we test directly with a *behavioural* proxy: hold out one known attack family, retrain JointCBM (γ=0) on the reduced known set, and keep the scorer (per-class Mahalanobis vs logit-space energy) that best detects the held-out family, then ask whether that choice matches the scorer actually best on the true unknowns (Mahalanobis on CTU-IoT-23, energy on CIC-IoT-2023). We repeat for every known attack family × five seeds (main Table VIII).

The proxy is only *conditionally* reliable. On CTU-IoT-23 the family-majority vote recovers Mahalanobis (two of three

TABLE S15: OOD scorer settings.

Scorer	Computation
Mahalanobis	Per-class means and a shared covariance of the within-class residuals ($+10^{-4}I$, pseudo-inverse), fitted on 20,000 training vectors; score = distance to the nearest class mean
Hamming, Jaccard	Hard rule vectors; each class representative is the majority vote of its training vectors (rule mean > 0.5, so it is binary); score = Hamming or Jaccard distance to the nearest representative
Energy	$-\log \sum_c e^{l_c}$ on the logits, T=1
Combined	Mahalanobis and energy scores each z-normalised with the mean and standard deviation of the known validation scores (frozen before testing), then summed
MSP	$1 - \max_c \mathrm{softmax}(l)_c$
ODIN	MSP at temperature T=1000 after an input perturbation of ϵ=0.0014 along the sign of the gradient
kNN	Euclidean distance to the 50th nearest L_2-normalised training embedding (20,000 training vectors)

families, $9/15$ runs); the C&C-HeartBeat and DDoS held-out-family experiments favour it, and only Okiru misfires (Mahalanobis cannot detect held-out Okiru, 0.22). On CIC-IoT-2023 the family-majority vote selects Mahalanobis for all four families ($15/20$ runs, $5/20$ energy) and so never recovers energy: held-out known families remain separable in activation space (Mahalanobis 0.70–0.86), so the proxy selects Mahalanobis, yet the true CIC unknowns overlap the known boundary and need energy. Held-out-known thus over-approximates activation-space detectability exactly on the corpus where the scorer must switch. In these experiments, therefore, this leave-family-out proxy did not consistently identify the best scorer; selecting it reliably would require observing the novel regime, which we do not demonstrate a priori. This is a further instance of the paper's negative result, here for scorer *selection* rather than scorer performance. Driver: `cbm/run_lofo.py`. A second rule that needs no retraining, based on synthetic outliers, is evaluated in Section SXV.

SXV. Transfer and Selection Tests

Do known-class statistics transfer across corpora? Main Table XIII reports within-corpus associations. To test prediction on an unseen corpus, we fit $\mathrm{AUROC} = a + b\,s$ by least squares on the 60 CBM concept-space models of two corpora, for each known-class statistic s, and apply it to the 30 models of the held-out third (Table S16). The prediction beats the training-corpora mean AUROC in 3 of 21 cases, ranks the held-out models in the right direction (positive Spearman correlation) in 7 of 21, and identifies the best configuration in 1 of 21. The fitted slope is dominated by the differences between corpora (the dataset means are non-monotone, main Section IV-D), so it often has the wrong sign even for CIC-IoT-2023 and TON_IoT, where the within-corpus direction agrees; statistics on different scales, such as ridge-LDA separability, also transfer poorly in absolute terms. Driver: `cbm/run_geometry_transfer.py`.

Scorer selection from synthetic outliers. The leave-family-out proxy (Section SXIV) needs retraining. As a second rule that uses only known traffic, we build synthetic outliers from known validation flows and select the scorer, per-class Mahalanobis or logit energy, that better separates them from a disjoint set of validation flows. The primary construction, fixed before evaluation, permutes each feature independently across flows, keeping every marginal and breaking the joint structure of real traffic; a secondary construction uses midpoints of random pairs of validation flows. The choice is compared, per model and seed, with the scorer that is better on the true unknowns, on the same evaluation subsets as main Table VII (Table S17). The permutation rule is right for $25/35$ CTU, $26/35$ CIC and $9/35$ TON_IoT models ($60/105$); the midpoint rule for $27/35$, $21/35$ and $9/35$ ($57/105$). Always choosing Mahalanobis is right for $62/105$ and always choosing energy for $43/105$, and the wrong choices of the permutation rule cost 0.066 AUROC on average. Both rules prefer Mahalanobis on TON_IoT, where energy is better for every architecture, so neither selects the scorer reliably. Driver: `cbm/run_scorer_selection.py`.

TABLE S16: Leave-one-corpus-out prediction of OOD AUROC from each known-class statistic (fitted on the other two corpora, 60 models; evaluated on the 30 held-out models). Slope: fitted b; MAE: mean absolute AUROC error on the held-out corpus; ρ: Spearman correlation of predicted and true AUROC; Best: the predicted best configuration is the true best.

Statistic	Slope	MAE	ρ	Best
Held out: CTU (training-mean MAE 0.167)				
Within-class variance	−1.774	0.198	−0.36	no
Fisher ratio	+0.065	0.169	−0.34	no
Ridge-LDA	+0.000	0.180	−0.38	no
Silhouette	+0.336	0.149	−0.34	no
Nearest-centroid margin	+0.201	0.152	−0.28	no
Concept entropy	+0.006	0.167	−0.38	no
Closed-set F1	−0.022	0.168	+0.17	yes
Held out: CIC (training-mean MAE 0.045)				
Within-class variance	+0.616	0.030	−0.66	no
Fisher ratio	+0.060	0.075	+0.60	no
Ridge-LDA	+0.003	0.671	+0.41	no
Silhouette	+0.263	0.069	+0.75	no
Nearest-centroid margin	+0.247	0.083	+0.54	no
Concept entropy	−0.415	0.064	−0.70	no
Closed-set F1	−0.006	0.045	−0.17	no
Held out: TON (training-mean MAE 0.123)				
Within-class variance	+2.001	0.146	−0.46	no
Fisher ratio	−0.045	0.145	−0.51	no
Ridge-LDA	−0.000	0.153	−0.58	no
Silhouette	−0.092	0.136	−0.43	no
Nearest-centroid margin	−0.113	0.149	−0.39	no
Concept entropy	+0.113	0.130	+0.37	no
Closed-set F1	+0.128	0.125	+0.30	no

SXVI. Open-Set (C+1) and Per-Family Evaluation

Protocol. We evaluate the deployed $(C{+}1)$-class detector at a reject threshold τ calibrated to 5% FPR on the known validation split. The scorer is per-class Mahalanobis on the model's scoring space: the concept activations for CBM variants, the binary rule activations for NeSy-NIDS, and the penultimate embedding for the MLP. Ground truth has C known classes plus a single "unknown" class into which *all* held-out families are merged; a flow is predicted "unknown" if its score exceeds τ, else the $\arg\max$ known-class logit. Evaluation uses 5,000 known test flows (all seeds) and, per unknown family, up to 3,000 flows: 9 families merged into "unknown" on CTU-IoT-23 and 29 on CIC-IoT-2023. Main Table IX reports, over five seeds, the known-class rejection rate

TABLE S17: Scorer selection from synthetic outliers (five seeds). True: OOD AUROC on the held-out unknown families; Perm.: AUROC separating validation flows from feature-permuted copies; Correct: seeds on which the rule picks the scorer that is better on the true unknowns (permutation / midpoint rule).

	True		Perm.		Correct	
Architecture	Mahal.	Energy	Mahal.	Energy	Perm.	Mid.
CTU-IoT-23						
MLP	0.879	0.386	0.984	0.766	5/5	4/5
JointCBM (γ=0)	0.830	0.476	0.802	0.703	4/5	4/5
JointCBM (γ=0.1)	0.812	0.494	0.795	0.747	3/5	4/5
JointCBM (γ=0.5)	0.851	0.570	0.823	0.747	4/5	5/5
JointCBM (γ=1.0)	0.842	0.612	0.853	0.786	5/5	5/5
SequentialCBM	0.875	0.604	0.644	0.585	4/5	4/5
HybridCBM	0.827	0.365	0.651	0.801	0/5	1/5
CIC-IoT-2023						
MLP	0.807	0.615	0.974	0.655	5/5	5/5
JointCBM (γ=0)	0.722	0.734	0.769	0.803	3/5	3/5
JointCBM (γ=0.1)	0.705	0.732	0.779	0.821	4/5	2/5
JointCBM (γ=0.5)	0.706	0.757	0.771	0.826	5/5	1/5
JointCBM (γ=1.0)	0.702	0.769	0.778	0.820	5/5	0/5
SequentialCBM	0.665	0.636	0.723	0.689	4/5	5/5
HybridCBM	0.670	0.629	0.678	0.732	0/5	5/5
TON_IoT						
MLP	0.550	0.662	0.859	0.523	0/5	0/5
JointCBM (γ=0)	0.644	0.713	0.740	0.621	1/5	0/5
JointCBM (γ=0.1)	0.640	0.674	0.758	0.624	3/5	2/5
JointCBM (γ=0.5)	0.641	0.694	0.770	0.630	2/5	2/5
JointCBM (γ=1.0)	0.642	0.706	0.758	0.637	1/5	1/5
SequentialCBM	0.603	0.640	0.730	0.637	0/5	2/5
HybridCBM	0.615	0.641	0.747	0.548	2/5	2/5

(target $\approx 5\%$), the open-set macro-F1 over the $C{+}1$ classes, and the equal-family-weighted mean AUROC.

The known rejection rate is close to target (4.2–5.8%). Across seeds the coefficient of variation of τ is at most 0.11 for every model (0.09 and 0.11 for NeSy-NIDS on CTU-IoT-23 and CIC-IoT-2023). Open-set macro-F1 is modest on CTU-IoT-23 (0.49–0.79) and low on CIC-IoT-2023 (0.13–0.27). The mixture is essential to reading this number. The evaluation pool is heavily unknown-skewed, for NeSy-NIDS on CIC-IoT-2023 (Table S18, seed 0) it is 5,000 known against 84,514 unknown flows, about 16.9 unknowns per known flow, and only 15,646 (18.5%) of the pooled unknown flows exceed τ. Macro-F1 weights every class equally, so it is *not* dominated by the unknown class; in that matrix the unknown class in fact has the *highest* per-class F1 (≈ 0.31). What pulls macro-F1 down is a prevalence effect on the *known* classes: the $\sim$81% of unknown flows that are not rejected are absorbed into known classes (most often VulnerabilityScan, Mirai, and DDoS-SYN), and because unknowns outnumber knowns $\sim$17:1 these false positives collapse known-class *precision* (Benign precision ≈ 0.07, so F1 ≈ 0.13) even though known *recall* stays high for accepted flows. The low unknown recall is a genuine weakness, but the small macro-F1 is as much a prevalence artefact as a detection failure: reweighting the same confusion matrix to equal known/unknown totals, with all conditional prediction rates unchanged, raises macro-F1 from ≈ 0.14 to ≈ 0.51, an arithmetic sensitivity check, not a better model. We therefore treat the operating-point macro-F1 as prevalence-sensitive and base the detection claim on the (prevalence-free) per-family AUROC of Table S19.

TABLE S18: ($C{+}1$) confusion matrix, NeSy-NIDS on CIC-IoT-2023 (seed 0; row = true, column = predicted). All 29 held-out families are merged into the single "unknown/reject" row. VulnSc.: VulnerabilityScan.

true \ pred	Benign	DDoS-SYN	Mirai	Recon	VulnSc.	reject
Benign	765	0	0	70	161	1
DDoS-SYN	0	911	0	0	0	37
Mirai	0	0	1073	0	0	3
Recon-PortScan	155	0	0	349	265	175
VulnScan	126	0	0	95	801	13
unknown	9633	13459	16918	6349	22509	15646

Table S19 shows per-family AUROC and unknown recall at τ for JointCBM (γ=0) and NeSy-NIDS; averaging its recall column over families with equal weight gives the family-averaged recall of main Table IX (0.51/0.36 for JointCBM, 0.83/0.20 for NeSy-NIDS on CTU/CIC). Pooled AUROC masks extreme per-family variation: for JointCBM detection is near-perfect for several volumetric DDoS families (AUROC > 0.97, recall > 0.8) but below chance for others (CIC Mirai-greip_flood 0.15; DDoS-UDP_Fragmentation 0.24; DDoS-SynonymousIP 0.48) and weak for most injection/scan/web families (0.58–0.71, recall $\leq$0.22), and NeSy-NIDS catches a different subset (Table S19). The equal-family-weighted mean AUROC (0.818 CTU, 0.722 CIC) is close to the pooled AUROC of main Table III (0.823 CTU, 0.717 CIC), but per-family detection ranges from near-perfect to near-chance for both models, so the pooled figure should not be read as a uniform per-family detection rate.

SXVII. Per-Class F1 on the Random Split

A single weighted F1 can hide minority-class failure. Table S20 gives per-class F1 on the known test split (random split, mean over five seeds) for the MLP and JointCBM (γ=0). The two agree within 0.002 for every class, so under matched-distribution evaluation this model shows no measurable per-class difference from the MLP; the other variants are not tabulated here. The weighted F1 nonetheless masks real spread: CTU Okiru is 0.86 and DDoS 0.90 (vs 0.98–1.00 for Benign and C&C-HeartBeat) and CIC Recon-PortScan is only 0.64, VulnerabilityScan 0.70 and Benign 0.76.

SXVIII. Joint (λ, γ) Sweep

λ (BCE concept supervision) and γ (added L2 concept fidelity) both push the predicted concepts toward the same labels, so we sweep them jointly (JointCBM, five seeds per cell, main splits, evaluation flows of main Table III; the λ=0.5 row is the JointCBM rows of that table) rather than as two 1-D slices (Table S21). Three points hold in 2-D. (i) γ is not a reliable lever: its effect $\gamma_{1.0}-\gamma_0$ is at most 0.040 and changes with λ, growing from -0.004 at λ=0.1 to $+0.030$ at λ=1.0 on CTU-IoT-23 and shrinking from -0.040 to -0.003 on CIC-IoT-2023. (ii) λ moves AUROC by a similar amount in

TABLE S19: Per-family OOD AUROC and unknown recall at τ (5% FPR) for JointCBM (γ=0) and NeSy-NIDS, mean over five seeds. n: evaluated flows of the family, the same for every seed (sampled without replacement, at most 3,000; each a distinct dataset record), so the five seeds do not add test flows and rare CTU families rest on very few flows. Detection varies enormously by family for both models, and the two disagree on which families they catch.

		JointCBM		NeSy-NIDS	
Held-out family	n	AUROC	Rec.	AUROC	Rec.
CTU-IoT-23					
C&C	3,000	0.832	0.33	0.864	0.29
Attack	3,000	0.812	0.45	0.995	0.99
C&C-HeartBeat-Attack	834	0.819	0.55	0.993	0.94
C&C-FileDownload	53	0.712	0.42	0.996	1.00
C&C-Torii	30	0.812	0.27	0.844	0.28
C&C-Mirai	2	0.994	1.00	0.994	1.00
C&C-HeartBeat-FileDownload	11	0.701	0.40	0.995	1.00
FileDownload	18	0.715	0.40	0.995	1.00
Okiru-Attack	3	0.964	0.80	0.979	0.93
CIC-IoT-2023					
DDoS-UDP_Flood	3,000	0.978	0.847	0.888	0.453
DDoS-TCP_Flood	3,000	0.902	0.390	0.830	0.107
DDoS-ICMP_Flood	3,000	0.949	0.638	0.900	0.481
DDoS-RSTFINFlood	3,000	0.974	0.816	0.993	0.958
DDoS-PSHACK_Flood	3,000	0.993	0.993	0.848	0.178
DDoS-SynonymousIP_Flood	3,000	0.475	0.002	0.350	0.015
DDoS-ACK_Fragmentation	3,000	0.962	0.821	0.516	0.055
DDoS-UDP_Fragmentation	3,000	0.237	0.024	0.892	0.343
DDoS-HTTP_Flood	3,000	0.969	0.881	0.868	0.606
DDoS-SlowLoris	3,000	0.977	0.898	0.705	0.284
DDoS-ICMP_Fragmentation	3,000	0.636	0.219	0.459	0.007
DoS-SYN_Flood	3,000	0.543	0.088	0.452	0.124
DoS-UDP_Flood	3,000	0.970	0.761	0.839	0.365
DoS-TCP_Flood	3,000	0.883	0.333	0.809	0.077
DoS-HTTP_Flood	3,000	0.975	0.907	0.897	0.697
Mirai-greip_flood	3,000	0.148	0.021	0.551	0.005
Mirai-udpplain	3,000	0.632	0.029	0.937	0.628
Recon-HostDiscovery	3,000	0.711	0.223	0.560	0.098
Recon-OSScan	3,000	0.661	0.066	0.537	0.074
Recon-PingSweep	2,262	0.635	0.091	0.454	0.014
MITM-ArpSpoofing	3,000	0.742	0.245	0.585	0.072
DNS_Spoofing	3,000	0.721	0.291	0.547	0.060
DictionaryBruteForce	3,000	0.582	0.065	0.441	0.007
BrowserHijacking	3,000	0.663	0.117	0.496	0.026
CommandInjection	3,000	0.647	0.156	0.487	0.008
XSS	3,000	0.597	0.103	0.441	0.006
SqlInjection	3,000	0.577	0.077	0.421	0.003
Backdoor_Malware	3,000	0.607	0.120	0.446	0.006
Uploading_Attack	1,252	0.596	0.128	0.431	0.004

opposite directions on the two corpora (down the γ=0 column, $+0.020$ on CTU, -0.040 on CIC). (iii) Known-class F1 is nearly constant across the grid (0.816–0.821 on CIC; 0.932–0.933 on CTU) except for two CTU runs at λ=1.0 (γ=0 seed 1 and γ=1.0 seed 2), where F1 falls to 0.734 and 0.736; both are *included* in the cell means, and they show that strong concept supervision can destabilise training. No single tunable factor governs open-set detectability.

SXIX. Concept-Fidelity Intervention (CBM)

Comparing soft concepts with their binarised values cannot establish semantic fidelity, since both can contain concept errors. Because our concepts are deterministic functions of the input, we test directly how much the JointCBM's accuracy depends on *deviations* from the declared predicates. We (i) run the model normally ($F1_{pred}$), (ii) intervene by replacing the predicted concepts with the *true* binary predicates $\mathbf{c}_{gt}$ and passing them through the model's own learned classifier head ($F1_{gt}$), and (iii) fit a fresh logistic regression on $\mathbf{c}_{gt}$ alone ($F1_{LR}$, how well this linear classifier can classify from the declared predicates; it is not an upper bound on the information they contain).

TABLE S20: Per-class known-test F1 on the random split (mean over five seeds). MLP vs JointCBM (γ=0); values agree to within 0.002.

Dataset	Class	F1 (MLP / JointCBM)
CTU	Benign	0.977 / 0.976
CTU	C&C-HeartBeat	0.999 / 0.998
CTU	DDoS	0.900 / 0.900
CTU	Okiru	0.857 / 0.857
CIC	Benign	0.760 / 0.759
CIC	DDoS-SYN_Flood	0.999 / 0.999
CIC	Mirai-greeth_flood	0.999 / 0.999
CIC	Recon-PortScan	0.636 / 0.638
CIC	VulnerabilityScan	0.699 / 0.697

TABLE S21: Joint (λ, γ) sweep: JointCBM OOD AUROC on the main splits (mean over five seeds; std in parentheses). λ down rows, γ across columns.

	γ=0	γ=0.5	γ=1.0
CTU-IoT-23			
λ=0.1	0.810 (.021)	0.801 (.040)	0.806 (.047)
λ=0.5	0.823 (.044)	0.844 (.036)	0.834 (.050)
λ=1.0	0.830 (.041)	0.854 (.029)	0.860 (.037)
CIC-IoT-2023			
λ=0.1	0.743 (.018)	0.724 (.023)	0.703 (.020)
λ=0.5	0.717 (.017)	0.702 (.007)	0.699 (.013)
λ=1.0	0.703 (.013)	0.704 (.005)	0.700 (.007)

The gap is large (Table S22). On CTU-IoT-23 the model scores F1 0.93 but its classifier on the true predicates scores only 0.46–0.49, and a logistic regression on the declared predicates alone reaches 0.58; on CIC-IoT-2023 the model scores 0.82 versus 0.62–0.65 from true predicates and 0.72 from a logistic regression on the predicates alone. So the eight declared predicates do not, under the evaluated linear classifiers, account for the CBM's known-class accuracy. Because the intervention alters the classifier's input distribution, this gap tests decision fidelity under predicate replacement; it does not attribute a fixed fraction of accuracy to a source. What it shows is that the learned concept values, whether through their continuous magnitudes or through their disagreements with the declared labels, carry decision-relevant information that the declared predicates do not. On CTU-IoT-23 the soft-versus-thresholded probe difference is small for $\gamma \leq 0.5$ (Section SII), so there most of that information survives thresholding. The mean gap shrinks slightly with γ (CTU $0.476 \rightarrow 0.441$), but by about one seed std (0.02–0.04).

TABLE S22: Concept-fidelity intervention (JointCBM, five seeds). $F1_{pred}$: normal; $F1_{gt}$: true predicates through the learned classifier; $F1_{LR}$: logistic regression on true predicates alone (deterministic, so no seed std). Values are mean (std) over five seeds. The intervention changes the classifier's input distribution, so the gap tests decision fidelity under predicate replacement, not a decomposition of accuracy.

Data	γ	$F1_{pred}$	$F1_{gt}$	gap
CTU	0	0.933 (.000)	0.457 (.024)	0.476 (.024)
CTU	0.1	0.933 (.000)	0.489 (.029)	0.444 (.029)
CTU	0.5	0.933 (.000)	0.483 (.016)	0.450 (.015)
CTU	1.0	0.933 (.000)	0.492 (.038)	0.441 (.038)
declared predicates alone (logistic regression): F1 $= 0.575$				
CIC	0	0.818 (.001)	0.624 (.021)	0.194 (.021)
CIC	0.1	0.819 (.001)	0.642 (.040)	0.177 (.040)
CIC	0.5	0.818 (.001)	0.645 (.028)	0.173 (.028)
CIC	1.0	0.818 (.001)	0.639 (.035)	0.179 (.036)
declared predicates alone (logistic regression): F1 $= 0.718$				